YOGESH O K

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EDUCATION

Bachelor of Engineering, Computer Science and Engineering Expected 2028
Chennai Institute of Technology, Chennai **CGPA: 7.98**
Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, System Design

PROJECTS

RepoLens | FastAPI, React, TypeScript, PostgreSQL, pgvector, Redis, Celery, AST, RAG, Docker GitHub

- Built a codebase-intelligence engine that pins each Git snapshot to an exact commit and extracts modules, classes, functions, methods, imports, calls and exact line ranges using Python AST parsing.
- Combined pgvector cosine retrieval with bounded dependency-graph expansion to answer architecture questions with file and line citations taken from the index, not generated by a model.
- Ran ingestion in Celery workers with Redis progress streaming, atomic re-indexing and tenant quotas; evaluated on 10 public repositories (640 files, 14.8k chunks): 75% recall@8 on 36 questions, 14 ms p95 queries.

Deploy-Lite | FastAPI, Docker, Celery, Redis, PostgreSQL, Nginx, MinIO, WebSockets, GitHub Webhooks GitHub

- Built a deployment engine that checks out the exact commit of a GitHub repository and builds frontend apps in isolated Docker containers with configurable memory, CPU, PID and timeout limits.
- Processed builds asynchronously with Celery and Redis, HMAC-verified idempotent GitHub webhooks, PostgreSQL deployment history and live WebSocket log streaming.
- Published build output to MinIO behind a single wildcard Nginx route for shareable per-deployment URLs; validated 10 simultaneous builds (all succeeded, 2.8 s p95) and 50/50 success, failure, redeploy and webhook workflows.

QueueLess | FastAPI, React, PostgreSQL, Redis, WebSockets, QR Workflows, Docker GitHub

- Built a real-time virtual queue with signed QR join, rotating-QR check-in, staff controls, cancellation, no-show handling and requeue, using FOR UPDATE SKIP LOCKED so concurrent calls never double-call a ticket.
- Pushed live position and wait-time updates over WebSockets with Redis pub/sub fan-out across API instances, and estimated waits with an exponentially weighted moving average of service times.
- Load-tested 300 participants (3 rounds of 100 simultaneous joins, 900 queue events): 80 ms p95 live-update latency and 13.5% median wait-time prediction error.

OpsFlow | FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Alembic, JWT, Redis, Docker GitHub

- Built a multi-tenant incident and release management platform for organizations to manage users, roles, teams, services, incidents, timelines, postmortems and releases with rollback.
- Implemented JWT authentication, level-based RBAC with roles re-read on every request, tenant-scoped authorization, Alembic migrations and single-transaction workflows across related resources.
- Added post-commit Redis events over WebSockets, Docker Compose and GitHub Actions CI with 105 automated tests, including a 72-case RBAC matrix and cross-tenant isolation checks.

INTERNSHIP EXPERIENCE

eGRCXperts - Software Development Intern | Chennai May 2026 - Jun 2026

- Built an RSA Archer workflow-automation system that transformed client workflow steps and Excel inputs into videos, mind maps, and UI visualizations for faster process understanding.
- Developed web-scraping and search-API pipelines with cosine-similarity classification to map cybersecurity-tool capabilities and support recommendation logic.
- Implemented API-oriented data transformation, automation, and structured-output workflows for internal GRC product analysis and engineering use cases.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript
Backend & Systems: FastAPI, REST APIs, WebSockets, Celery, JWT, RBAC, SQLAlchemy, Alembic, Redis
Frontend: React, React Router, HTML5, CSS3
Databases & Data: PostgreSQL, pgvector, MySQL, MinIO
DevOps & Tools: Docker, Docker Compose, Nginx, Git, GitHub Actions, CI/CD, Postman, pytest, Playwright
AI / Data: RAG, Embeddings, Abstract Syntax Trees (AST), Dependency Graphs, Web Scraping, Prompt Engineering, Cosine Similarity
Software Engineering & Core CS: System Design, Distributed Systems, API Design, Database Design, Unit/Integration Testing, DSA, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS & LANGUAGES

- LeetCode Knight with 900+ problems solved across core data structures, algorithms, graphs, and dynamic programming.
- English: Professional working proficiency.
- Japanese: JLPT N3 Qualified; currently studying for JLPT N2.
- Tamil: Native proficiency.